

RESEARCH LETTER

Changes in soil hydraulic conductivity in sweet potato field with living mulch

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Abstract

Living mulch (LM) is used in agricultural fields to suppress weeds, control diseases, and mitigate erosion. It also enhances soil nutrient supply at the root death and decay stage during the growing season. However, benefits of LM to soil hydraulic properties related to soil pore structure have not been elaborated here. We focus on temporal changes in soil hydraulic conductivity (K) in a field where sweet potato was grown with and without LM (barley, *Hordeum vulgare* L.). K was measured in the field using a mini-disk infiltrometer at three different pressure heads. In the plots with LM, K decreased significantly in August and then increased in October compared to plots without LM (at -0.5 cm pressure head). Changes in soil pore structure due to root growth or death and decay may alter soil hydraulic conductivity.

1 | INTRODUCTION

High rainfall and steep topography in Japan cause agricultural soils vulnerable to surface erosion (Mihara, 2001). Living mulches (LMs) and cover crops help reduce erosion by root reinforcement (Gyssels et al., 2005), mitigating rain-drop impact (Zuazo et al., 2008) and increasing infiltration by aggregate formation with C inputs (Siller et al., 2016). LMs are unharvested companion crops that support growth of primary crops during the growing season. LM is planted between vegetable rows and is used as winter green manure

input (Paine & Harrison, 1993). LMs also suppress weeds (Bhaskar et al., 2021), control diseases (Kosinski et al., 2021), and supply nutrients to soils (Qian et al., 2015).

Benefits of LM also include mitigating soil temperature changes (Liedgens, Soldati, et al., 2004) and reducing surface evaporation (Teasdale & Daughtry, 1993); however, little information exists on benefits LM provides to soil hydraulic properties affected by root growth induced changes in soil structure. Water, heat, and solute transport in soils support crop growth, and these are strongly related to soil pore structure (e.g., Athmann et al., 2013). As LM grows and dies,

Abbreviations: LM, living mulch.

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root development changes and so does the soil pore structure. Therefore, characteristics of flow and transport in soils may correspondingly change.

To address these gaps, we have investigated how LM affects soil hydraulic conductivity (K) during the growing season in a field with sweet potatoes (*Ipomoea batatas* L.) as a main crop. LM treatments are compared to plots without LM (hereafter known as controls). Sweet potatoes require more water during the summer growing season and dry conditions during fertilization before harvest enhances starch accumulation and taste. We hypothesized the following changes to soil structure during the sweet potato cultivation season: soil is compacted and K decreases during LM root growth, and K increases as macropores are formed during root death and decay.

2 | MATERIALS AND METHODS

2.1 | Experimental site and data treatment

The climate in this region is humid subtropical with monthly precipitation during the growing season (May to October 2019) ranging from 56.5 to 390.5 mm and mean monthly temperature ranging from 18.8°C to 27.9°C. The soil at this site is a typical andosol (Kanto loam) in the World Reference Base (Food and Agriculture Organization [FAO], 2015) with sandy loam texture (sand 66%, silt 28%, and clay 6%).

The experiment was conducted in a sweet potato field (variety “Beniharuka”) at the Center for International Field Agriculture Research and Education, Ibaraki University (36°03'N, 140°21'E), Japan. Three replications of sweet potato cultivation with and without LM were established in 3 m × 10 m plots. Barley (*Hordeum vulgare* L.) was used as LM and sowed on May 24, 2019 between and beside sweet potato rows; LM began senescing 50 days after sowing. The soil's physical measurements and samples were taken near margins of plot ridges and access was prevented to avoid trampling. Field measurements were conducted from June to October 2019 once in a month with three replicates of both LM and controls. Student's t -test was used to assess differences of means for measured soil properties.

2.2 | Soil hydraulic conductivity

In situ soil hydraulic conductivity (K) was measured monthly using a minidisk infiltrometer (METER Group, Inc.) at pressure heads of −0.5, −3 and −6 cm (Casey et al., 1998; Logsdon et al., 1993). Assessing soil hydraulic conductivity during infiltration requires measuring cumulative infiltration versus time and fitting results as follows (Zhang, 1997):

$$I = C_1 \sqrt{t} + C_2 t \quad (1)$$

Core Ideas

- Near-saturated hydraulic conductivity decreased in summer and increased in fall with LM.
- Temporal changes in soil pore structure due to LM root dynamics alter hydraulic conductivity.
- Barley as a living mulch benefits sweet potato crops by modifying water conditions in soil.

where C_1 relates to soil sorptivity (m s^{-1}) and C_2 (m s^{-1}) relates to hydraulic conductivity, and t is time (s). Hydraulic conductivity for the soil (K) is computed as

$$K = \frac{C_1}{A} \quad (2)$$

where A relates to van Genuchten parameters (n and α) of different soil textures for pressure head and disk radius:

$$A = \frac{11.65 (n^{0.1} - 1) \exp [2.92 (n - 1.9)] \alpha \psi_i}{(\alpha r)^{0.91}} \quad n \geq 1.9 \quad (3)$$

$n = 1.9$ and $\alpha = 0.08$ for our soil, r is disk radius (cm), and ψ_i is pressure head (cm) at the disk surface. A pressure head of −0.5 cm was used to measure near-saturated hydraulic conductivity, while −3.0 and −6.0 cm heads represented unsaturated hydraulic conductivities.

2.3 | Soil bulk density and readily available water content

Soil bulk density (ρ_b) and readily available water content for normal crop growth (θ_{RAW}) were measured in core samples (5.0 cm diameter; 5.0 cm long) collected from each plot in June and October. Soil core samples were collected at 5.0 cm intervals from the surface to a depth of 20 cm and θ_{RAW} was measured using a sand suction table and the pressure plate method. We calculated θ_{RAW} as the difference in volumetric water content corresponding to soil water pressure heads of −30 cm and −1000 cm (Hasegawa, 1997). After θ_{RAW} measurement, ρ_b was calculated for each sample by dividing oven-dry mass of soil (105°C for 24 h) by the soil core volume.

2.4 | Pore connectivity index

Pore connectivity index (C_w) was calculated considering the soil pore connection structure (Lozano et al., 2014):

$$C_w = \frac{\Delta K (\psi_i, \psi_{i+1})}{\Delta \theta (\psi_i, \psi_{i+1})} \quad (4)$$

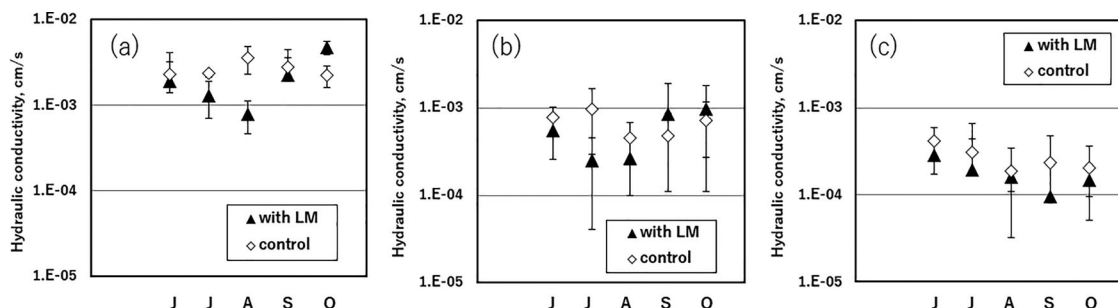


FIGURE 1 Soil hydraulic conductivity measured at (a) -0.5 cm, (b) -3.0 cm, and (c) -6.0 cm pressure heads. All data shown are arithmetic means of three replicates. Error bars are ± 1 SD. LM, living mulch.

where $\Delta K(\psi_i, \psi_{i+1})$ is the difference in K corresponding to soil water pressure heads ψ_i and ψ_{i+1} used in the infiltration test, and $\Delta\theta(\psi_i, \psi_{i+1})$ is the difference in soil volumetric water content (θ , cm^3). Each θ value was calculated from a soil water retention curve constructed at the corresponding value of ψ . Differences between pressure heads of -0.5 and -3.0 cm corresponding to pore radii $r_p > 5$ mm define macropores, whereas differences between heads of -3.0 and -6.0 cm (pore radii $5\text{-mm} > r_p > 2.5$ mm) define mesopores (Lozano et al., 2016).

3 | RESULTS AND DISCUSSION

3.1 | Soil hydraulic conductivity

Some differences in K occurred between LM plots and controls at -0.5 and -3.0 cm pressures during the mid- to late growing season (Figure 1). In the -0.5 cm pressure head test, K significantly decreased in August and increased in October in LM treatments compared to controls. At -6.0 cm pressure heads, differences in K between LM treatments and controls were not as clear. K values in LM plots tended to be smaller than controls during the early to mid-growing season, but this trend somewhat reversed for the -3.0 cm pressure head test in October.

We hypothesized that near-saturated K declined during summer because the roots of LM thrive and compress soil in the rhizosphere, while increases in near-saturated K in autumn may be attributed to decaying roots and concurrent formation of biopores. Observations using X-ray μCT images have shown soil compaction during root growth (Lucas et al., 2019a); thus, our summer results may reflect this phenomenon. Or et al. (2021) reported that higher saturated hydraulic conductivity and infiltration capacity may be attributed to cumulative effects of vegetation, particularly increases in biopores formed by decaying roots (Dunne et al., 1991; Lucas et al., 2019b; Thompson et al., 2010; Wuest, 2001). Because the -0.5 cm pressure head test represents

near-saturated conditions, our results in autumn may reflect this phenomenon. A disadvantage of LM is the competition for water usage such as transpiration, which has been reported when Italian ryegrass was used with soybeans as LM (Liedgens, Frossard, et al., 2004); however, no competition for water use in summer was observed in visual of sweet potato growth, likely because the barley grew to only ≈ 30 cm high and died standing.

3.2 | Soil bulk density and readily available water content for crops

Most changes in ρ_b measured between June and October are not significant (Figure 2a,b). Small reductions of ρ_b in surface depths of LM may reflect decay of surficial roots and formation of biopores. Since roots of most grain crops are about 15 cm long after 30 days of growth (National Agriculture and Food Research Organization [NARO], 2009), it is likely that the roots developed in shallow depths died by late summer or early fall forming biopores that decreased ρ_b in the surface soil. Similar, but slight, decrease in ρ_b of surface soil occurred in controls possibly due to roots of weeds that were not removed. The largest seasonal increase in ρ_b (12.5 cm depth in controls) may have resulted from soil compaction between rows due to lateral expansion of sweet potato roots. Minor differences noted in ρ_b could also be attributed to natural variability in soil properties. The slightly lower ρ_b in deeper layers (Figure 2a) might reflect soil compaction by precipitation or human traffic after tillage.

θ_{RAW} with and without LM was not significantly different except in the 0–5 cm layer in October ($p < 0.10$; Figure 2c,d). However, overall θ_{RAW} was larger in June and smaller in October in LM plots compared to controls, possibly related to roots of LM decreasing the connectivity of soil pores. Thus, water holding capacity may have increased somewhat in June and decreased in October due to biopore formation by dead and decayed roots, which increased K and decreased θ_{RAW} .

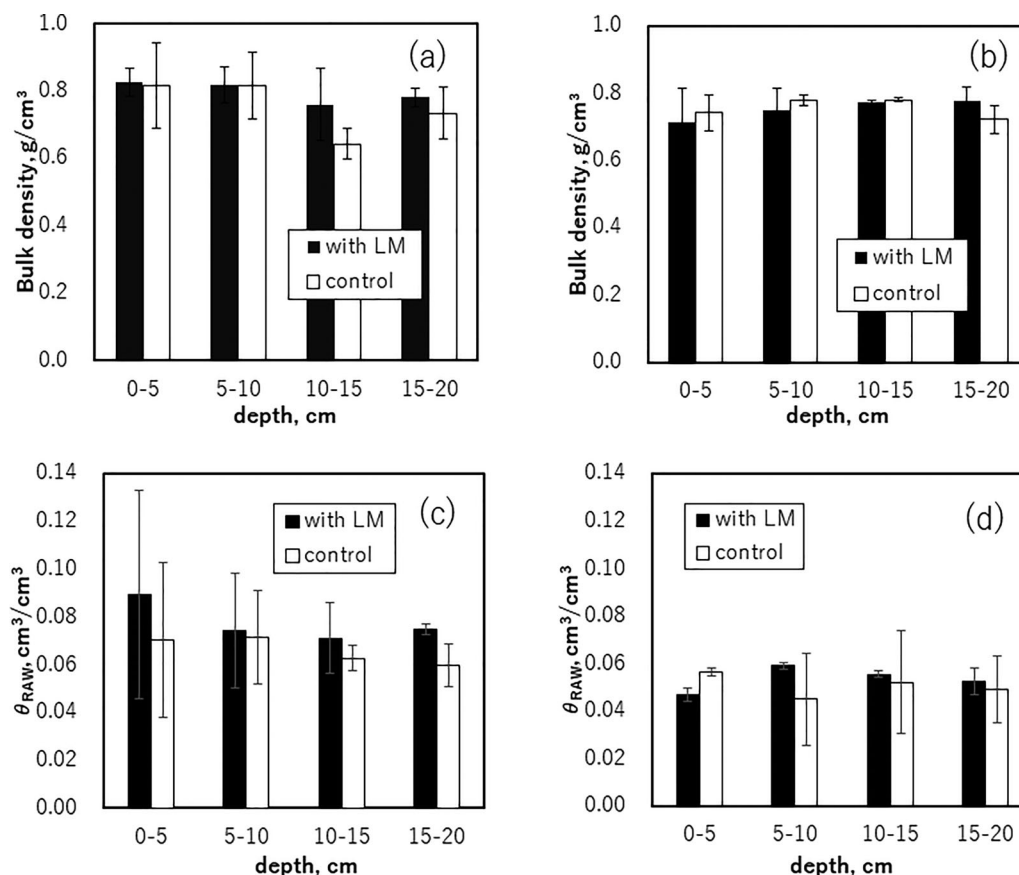


FIGURE 2 Bulk density (ρ_b) in (a) June and (b) October. Readily available water content (θ_{RAW}) in (c) June and (d) October. Error bars are ± 1 SD. LM, living mulch.

3.3 | Pore connectivity index

Pore connectivity index due to macropores significantly increased from June ($4.15 \pm 0.25 \text{ cm h}^{-1}$) to October ($11.37 \pm 5.15 \text{ cm h}^{-1}$) in plots with LM, while seasonal decline in pore connectivity was observed in controls ($7.13 \pm 7.52 \text{ cm h}^{-1}$ and $3.15 \pm 0.66 \text{ cm h}^{-1}$, respectively (Figure S1a)). Similar, but less pronounced, temporal patterns occurred for mesopores (Figure S1b). Plots with LM had increased in biopores caused by dead and decayed roots, as widely known (e.g., Gomi, 2016; Noguchi et al., 1999). In controls, decline in pore connectivity indices may be due to surface soil compaction caused by precipitation and weed root growth.

4 | CONCLUSIONS

Soil pore structure varied with the growth and death of LM roots causing changes in K during the sweet potato growing season. Specifically, K decreased in summer due to soil compaction affected by growth of LM roots and K increased in autumn because larger pores formed and increased pore

connectivity after LM roots died and decayed. The significant difference of θ_{RAW} between LM and control plots occurred only in 0–5 cm depth; however, θ_{RAW} was higher in early summer and lower in autumn. Sweet potato requires adequate water in the growing season and dry conditions are preferable at harvesting. Thus, our observations indicate that *Hordeum vulgare* L. benefited from these soil physical property changes related to soil water movement to sweet potato cultivation.

AUTHOR CONTRIBUTIONS

Junko Nishiwaki: Conceptualization, data curation, formal analysis, investigation, methodology, writing – original draft. **Takuya Koseki:** Investigation. **Naomi Asagi:** Investigation, methodology, resources. **Hirotaka Saito:** Funding acquisition, writing – review and editing. **Roy C. Sidle:** Methodology, writing – review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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SUPPORTING INFORMATION

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